

Stat 1040:

Solutions for

Suggested Review Exercises

1. (a) Too hasty. What about population size?

Comments: Michigan may include the big bad city, but Minnesota has twice the population of Michigan. The crime rate is lower in Michigan. (Of course, there probably are neighborhoods in Detroit that are best avoided.)

- (b) This is better reasoning, because the population of the U.S. increased over the period 1991–2001. Looking at rates would make the point even more clearly. (In fact, there has been a remarkable decline in crime rates over the period 1980–2005.)

4. (a) They were controlling for age and sex as possible confounders; this is discussed on p. 13, with respect to a specific disease—lung cancer.

- (b) This is the wrong conclusion to draw. Ex-smokers are a self-selected group, and many people give up smoking because they are sick. So recent ex-smokers include a lot of sick people. (Other epidemiological data show that if you quit smoking, you will live longer.)

5. No. The data from the double-blind study are more reliable, and suggest that the results from the single-blind were biased.

7. (a) This is an observational study, so confounding may be a problem.

- (b) Rates of cervical cancer go up with age. Women of different marital status have different patterns of sexual activity, and are therefore exposed to different kinds of risk; similarly for education. In other words, age, marital status, and education are potential confounders.

- (c) Pill users are more active sexually than non-users, and have more partners. That seems to be what makes the rate of cervical cancer higher among pill users. (This is like example 2 on p. 16 or exercise 11 on p. 23.)

- (d) No; see (c).

9. (a) False. (b) True. (c) False: that is the whole point of experiments.

Discussion. People who eat lots of fruits and vegetables are different from the rest of us in many other ways. Some other aspect of diet or life style may be protective. Of course, the observational studies might be right; something in the fruits and vegetables other than the vitamins might be the protective factor.

10. (a) Observational study. (b) Yes. (c) Yes.

- (d) No. The gene would also have to be associated with controlling behavior by the mother (p. 20).

- (e) A mother who sees her child eat too much might respond in a way that psychologists would interpret as “controlling”—Johnny, stop eating!

- (f) No. The *Chronicle* seems to have over-reacted.

11. (a) The treatment group consists of those who finished boot camp. The control group consists of other prisoners—including those who do not volunteer, or those who volunteer but do not complete the program.

- (b) This is observational. The prisoners decide whether to volunteer for boot camp and whether to stay in the program or drop out. That is the problem: those who volunteer and stay the course might be quite different from who volunteer but drop out.

- (c) False.

Comment. An experiment could be done either like the polio trial (p. 1ff) or the HIP trial (exercise 9 on pp. 22–23):

Like the polio trial. Ask for volunteers. Randomize some of the volunteers to treatment (assignment to boot camp) and some to control. Compare the recidivism rate for the two groups—but include the dropouts in the treatment group. (Otherwise, you still have the problem of self-selection.)

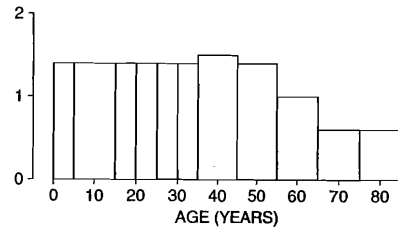
Like the HIP trial. Take a group of prisoners. Randomize some to treatment (invitation to participate in boot camp) and some to control. Compare the recidivism rate for the two groups—but include in the treatment group those who decline to participate and those who drop out. (Again, this is to guard against the problem of self-selection.)

12. False. The conclusion does not follow. This is just like the admissions study (pp. 17ff). The Democrats may be concentrated in wards with low turnouts. Here is an example, with only two wards (and see exercise 13 on p. 24).

	DEMOCRATS		REPUBLICANS	
	Total number	Number voting	Total number	Number voting
Ward A	1000	100	100	5
Ward B	100	60	1000	500

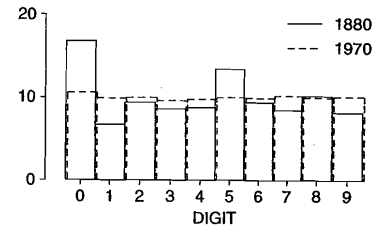
2. (a) There are more at age 1. The histogram is higher at 1 than at 71.
 (b) There are more at age 21.
 (c) There are more age 0–4,
 (d) 50%

Histogram for review exercise 2, chapter 3



4. (a) 25% (b) 99% (c) 140–150 mm (d) 135–140 mm
 (e) About $5 \times 2.1 = 10.5\%$ (f) 102–103 mm
 (g) 117–118 mm is a good guess; the interval is somewhere between 115 and 120 mm.

6. (i) and (ii) not (iii).
 Reason: With lists (i) and (ii), 25% of the people have heights between 66.5 inches and 67.5 inches; 50% between 67.5 and 68.5 inches; 25% between 68.5 and 69.5 inches. Not so with list (iii).
 10. (a) The histogram is shown below.



- (b) In 1880, people did not know their ages at all accurately, and rounded off.
 (c) In 1970, people knew when they were born.
 (d) In 1880, there was a strong preference for even digits (although 4 and 6 lose out to 5); again, this is probably due to rounding. In 1970, the preference was much weaker.

1. (a) Average = 50. Deviations = -9, -2, 0, 0, 4, 7. The SD is

$$\sqrt{\frac{(-9)^2 + (-2)^2 + 0^2 + 0^2 + 4^2 + 7^2}{6}} = 5.$$

- (b) 48, 50, 50 are within 0.5 SDs of average, i.e., in the range from 47.5 to 52.5.
48, 50, 50, 54, 57 are within 1.5 SDs of average, i.e., in the range from 42.5 to 57.5.
3. (a) 5. The average should be in the middle of the distribution: only three of the numbers are smaller than 1, and none are bigger than 10.
- (b) 3. If the SD is 1, the entries 0.6 and 9.9 are much too far from average. The SD can't be 6, because none of the numbers are more than 6 away from the average.

6. (a) (i) 60 (ii) 50 (iii) 40

- (b) (i) median is bigger than the average—long left hand tail
(ii) median is about equal to the average—symmetry
(iii) median is less than the average—long right hand tail
- (c) 15. Most of the area is within 50 of the average, so 50 is too big. Only a little of the area is within 5 of the average, so 5 is too small.
- (d) False. Histograms (i) and (iii) are almost mirror images, and have just about the same SD.

7. (a) Average weight of men = $66 \times 2.2 \approx 145$ pounds, SD ≈ 20 pounds.
Average weight of women ≈ 121 pounds, SD ≈ 20 pounds.

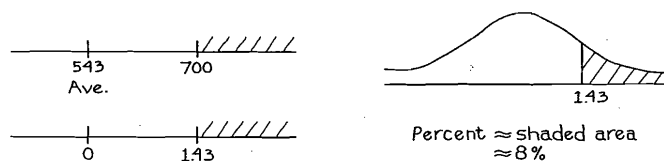
- (b) 68%: the range is average ± 1 SD.
- (c) bigger than 9 kg: if you take the men and the women together, the spread in weights goes up.

1. (a) 79% of 25 $\approx$ 20.



- (b) 18.

3. (a) In 1967, a score of 700 was 1.43 SDs above average: $(700 - 543)/110 \approx 1.43$. The percentage scoring over 700 was about 8%—



- (b) In 1994, a score of 700 was 1.83 SDs above average, and the percentage scoring over 700 was about 3%.

4. (a) 9% (b) 5%

5. The percentage of men with heights between 66 inches and 72 inches is exactly equal to the area between 66 inches and 72 inches under the histogram. This percentage is approximately equal to the area between -1 and +1 under the normal curve.

7. (a) 400 is 1.5 SDs below average; this student is in the 7th percentile of the score distribution (approximately), because the area to left of -1.5 under the normal curve is 7%.

- (b) The student needs to be about 0.7 SDs above average, so, needs a score of about 620 on the M-SAT. (The 75th percentile is nearly 0.675, rounded here to 0.7.)

8. (a) True. See pp. 92ff.

- (b) False: all the deviations from average stay the same.

- (c) True.

- (d) True: all the deviations from average are doubled.

- (e) True.

- (f) False: all the deviations from average have their signs changed, but that goes away in the squaring. The SD has to be positive (or, exceptionally, zero).

9. (a) False. For example, the list 1, 2, 99 has a median of 2 and an average of 34.

- (b) False. For the list 1, 2, 99, two-thirds of the entries are below average.

- (c) False. For example, income data (p. 88) have a long right-hand tail; educational levels have a long left-hand tail as well as bumps at 8, 12 and 16 years (p. 39).

- (d) False. If the histogram for a list follows the normal curve, the percentage of entries within 1 SD of average will be around 68%. Otherwise, the percentage could be a lot different. For instance, take a list of eight numbers: one is 30, one is 70, and six are 50. The average is 50, the SD is 10, and 75% are within 1 SD of average.

4. 0.03 inches or so.

§ 6. (a) $(600 + 650)/2 = 625$

(b) More than 125. When you put the men and women together, the spread goes up, because the two distributions are different.

§ 8. You would be too low, because the curve is lower than the histogram in that range.

§ 10. False. The data are cross sectional not longitudinal. The 20-year-olds were born 1956–60; the 70-year-olds, 1906–10. In the early part of the century, there was much more pressure to conform and be right-handed.

Comment. Some investigators believe that left-handed people suffer higher mortality, and use the trend in left-handedness as supporting evidence. See Stanley Coren, “Left-handedness and accident-related injury risk” *American Journal of Public Health* vol. 79 (1989) pp. 1040–41, and “The diminished number of older left-handers: differential mortality or social-historical trend?” *International Journal of Neuroscience* vol. 75 (1994) pp. 1–8. However, Coren’s argument seems a little shaky. For some evidence in the other direction, see M. E. Salive et al., “Left-handedness and mortality,” *American Journal of Public Health* vol. 83 (1993) pp. 265–7.

§ 13. (a) Heart disease rates go up with age: the drivers could be older, accounting for the difference in rates. That explanation has been eliminated.

(b) You want the difference in rates to be due to exercise on the job, rather than pre-existing factors. It might take some time for exercise to have its beneficial effect.

(c) Drivers and conductors are going to be similar with respect to age, education, income, and so forth.

(d) This is an observational study, so there may be some confounding. Drivers may be more at risk than conductors to start with. For example, drivers may be heavier.

(e) You could look to see if the drivers and conductors had similar body sizes when they were hired; that would suggest the two groups were comparable at time of hire, and strengthen the argument that exercise matters. If drivers were heavier than conductors at time of hire, the argument for exercise is weaker.

Comment. As it turned out, the drivers were heavier than the conductors at time of hire, explaining the difference in rates of heart disease.

Answers to Review Exercises: Chapter 8

1. The answer is (d). With (a), the averages are too low. With (b), the SDs are too small. With (c), the SDs are too big and the correlation is too high.
2. (a) Negative: older cars get fewer miles per gallon.

(b) Richer people own newer cars, and maintain them better. (Some rich people own Ferrari gas guzzlers, but not many; and 10-year-old Chevrolets in poor repair might guzzle even more.)
4. 0.3. Taller men do marry taller women, on average. But there is lots of variation around the line.
11. $r = -1$: number wrong = $10 - \text{number right}$, so all the points on a scatter diagram (for number wrong vs. number right) lie on a straight line which slopes down. Also see exercise 9 on p. 106.

Answers to Review Exercises: Chapter 9

2. (a) False. The scatter diagram slopes down: if x is below average, y is generally above average.
(b) False. For example, height at age 9 is usually less than height at age 18, but the correlation is positive.
8. False. The data are cross-sectional, not longitudinal. Younger people were born later and educational levels have been going up over time.
11. Quite a bit less. See section 4, and exercise set B on pp. 145ff.
12. (a) Education is measured in whole years (years of schooling completed).
(b) Some dots correspond to many couples. For example, in many families, the husband and the wife both have 12 years of education (high school degree). All such families are plotted on the (12, 12) dot.
(c) A (iv) B (iii) C (i)

Answers to Review Exercises: Chapter 10

1. A (i) B (iii) C (ii)
2. (a) 112. Work: 115 is 1 SD above average at age 18, so the estimate at age 35 is above average by r SDs. This is $0.8 \times 15 = 12$ points.
(b) 112. See section 3.
3. (a) 64 inches (b) 62 inches (c) 63 inches, the average (d) same as (c)
Work for (a). The husband is 4 inches, or $4/2.7 \approx 1.5$ SDs above average in height. The wife is predicted to be above average in height by
$$r \times 1.5 = 0.25 \times 1.5 \approx 0.4 \text{ SDs.}$$
This is $0.4 \times 2.5 = 1$ inch.
4. (a) 15 years.
(b) 13.5 years.
(c) Appearances are deceiving; all that is going on here is the regression effect (section 5).

Answers to Review Exercises: Chapter 11

1. Option (v): see p.186.
2. Something is wrong. GPAs run from 0 to 4. If you predict 2, the maximum error is 2. The computer should be doing better than that.
3. (a) $\sqrt{1 - 0.8^2} \times 2.5 = 1.5$ inches.
(b) $\sqrt{1 - 0.8^2} \times 1.7 \approx 1.0$ inches.
4. (a) r.m.s. error = $\sqrt{1 - r^2} \times \text{SD of final scores} = 12$.
(b) 65.8. This student is $30/25 = 1.2$ SDs above average on the midterm, and should be above average on the final by $r \times 1.2 = 0.72$ SDs. That is 10.8 points.
(c) 12, the r.m.s. error: see part (a). It is OK to use the r.m.s. error inside a strip because the diagram is football-shaped.
6. No. The conclusion seems right, but does not follow from the data. It could be, for example, that better students spend more time doing homework anyway.
7. Option (iii) is right: regression effect.
12. True, although the difference is pretty small. The correlation is negative: the men with 20 years of education should have blood pressures which are, on average, below the grand average by about 3 mm (by the regression method). In other words, the men with 20 years of education should have average blood pressures of about 116 mm.

Answers to Review Exercises: Chapter 12

2. Predicted income is $(\$1600 \text{ per inch}) \times \text{height} - \$81,400$. Taller people make more money, on average. Probably, this reflects other variables in family background; although looking every inch an executive may not hurt.
3. Obviously not. The slope means that the 151-pound men are taller, on average, than the 150-pound men—by around 0.0267 inches. Similarly, the men who weigh 152 pounds are on average a little taller than the men who weigh 151 pounds. And so forth.
5. This is not legitimate. There are two regression lines. The one for predicting husbands from wives has slope 0.375. See section 10.5.
8. (a) True. (b) False. (c) False. (d) True. (e) True.
Comment. See section 12.2.
9. This is the regression line of IQ on parental income—in disguise (section 10.2). The slope is
$$r \times \text{SD of IQ} / \text{SD of income} = 0.50 \times 15 / 45,000 = 1/6000.$$
10. \$150,000 is likely to be too high. You need the other regression line, for income on IQ (section 10.5). A better estimate is \$105,000.
11. Something is wrong with the equation. The regression line has to go through the point of averages. This line doesn't.
12. No. There doesn't even seem to be any consistent direction to the association.

Answers to Review Exercises: Chapter 13

2. Option (i) is better: you only have to jump one hurdle rather than two.

3. Option (ii) is better, because there are a lot more ways to win:

clubs diamonds hearts spades
hearts diamonds clubs spades
etc.

6. (a) True.

(b) True. See example 2 on p.226.

(c) False. The two events are dependent. See example 4 on p.229. The chance is $1/52 \times 1/51$, not $1/52 \times 1/52$.

7. (a) False. (b) False. (c) True.

Both sequences are equally likely, having chance $1/2^6$. Of course, there are a lot more sequences with 3 heads and 3 tails, but that's not the question.

8. (a) On one roll, the chance of getting 3 or more spots is $4/6 \approx 0.67$; the chance of getting 3 or more spots on 4 rolls is $(0.67)^4 \approx 0.20$, or 20%.

(b) None of the rolls show 3 or more spots if all show 2 or fewer, and the chance is $(2/6)^4 \approx 0.012$, or 1.2%.

(c) $100\% - 20\% = 80\%$.

Comment. Compare the two propositions,

(i) "not all the rolls show 3 or more spots," and

(ii) "none of the rolls show 3 or more spots."

Proposition (i) is easier, and has a bigger chance. For example, suppose the dice land 4 6 5 2. Then (i) holds—the last roll was less than 3. But (ii) does not—the first roll is bigger than 3. (So were the second and third rolls.)

9. (a) $(1/6)^{10} = 1/60,466,176$.

(b) $1 - 1/60,466,176 \approx 1$.

(c) $(5/6)^{10} \approx 0.16$, or 16%.

Comment. Not getting 10 sixes is easier than getting 10 non-sixes.

10. Option (ii) is better. You have the same 50–50 chance of winning \$1, but you can't lose.

12. $3/100 \times 2/99 \times 1/98 \approx 6/1,000,000$.

1. (a) The chance is $1/6 \times 1/6 = 1/36$.
 (b) The chance is $6/36 = 1/6$; see figure 1 in chapter 14. Or use part (a) and the addition rule. Here is yet another argument. Imagine one of the dice is white and the other is black: no matter how the white one lands, the black one has 1 chance in 6 to match it.
4. Option (i) is better. Even if you miss the first time, you get a second try at the money.
6. If you want to find the chance that at least one of the two events will happen, check to see if they are mutually exclusive; if so, you can add the chances.
 If you want to find the chance that both events will happen, check to see if they are independent; if so, you can multiply the chances.
7. This is like Chevalier de Méré: the chance is $1 - (3/5)^4 \approx 87\%$.
8. 100%—you can't avoid it.

9. Draw a picture like figure 1—

1 1	1 2	1 3	1 4
2 1	2 2	2 3	2 4
3 1	3 2	3 3	3 4

For instance, 3 2 means you got 3 from box A and 2 from box B. There are 12 outcomes, and each has chance $1/12$.

- (a) The good outcomes are 2 1, 3 1, 3 2; the chance is $3/12 = 25\%$.
- (b) $3/12 = 25\%$.
- (c) $100\% - (25\% + 25\%) = 50\%$.
10. Option (ii) is better. There are 60 rolls, and 60 draws. On each roll you have 2 chances in 6 to win \$1 and 4 chances in 6 to get nothing; but on each draw, you have 3 chances in 6 to win \$1 and 3 chances in 6 to win nothing.
11. (a) $13/52 \times 12/51 \times 11/50 \approx 1\%$.
 (b) $39/52 \times 38/51 \times 37/50 \approx 41\%$.
 (c) The chance of getting all diamonds is 1%, see (a). The chance of not getting all diamonds is 99%.

Comment. "No diamonds" and "not all diamonds" are two different propositions.

- § 1. False. The population got bigger too. You need to look at the number of murders relative to total population size. The population in 1990 was about 249 million, and in 1970 it was about 203 million: 20,273 out of 249 million is actually a bit less than 16,848 out of 203 million.

Comment. The murder rate declined sharply from 1990 to 2000.

- § 3. False. The selection ratio for the pooled data is 68.5%. This is like Simpson's Paradox (pp. 17ff). A majority of the women took the test in 1977, when the pass rates were higher.

Technical comment. In the pooled data, the women's pass rate is a weighted average of their pass rates for 1975 and 1977, and must be intermediate between those two values. Likewise for the men. The selection ratio, however, is a more complicated statistic. The selection ratio in the pooled data can be written as

$$\frac{79 \times \frac{10}{79} + 102 \times \frac{18}{102}}{79 + 102} \bigg/ \frac{1312 \times \frac{250}{1312} + 1259 \times \frac{331}{1259}}{1312 + 1259}$$

which equals

$$\frac{79 \times 0.1266 + 102 \times 0.1765}{79 + 102} \bigg/ \frac{1312 \times 0.1905 + 1259 \times 0.2629}{1312 + 1259}.$$

The women are in the numerator; the men, in the denominator. The weights for the women and for the men differ. The women get more weight on the higher pass rate (102 on 0.1765 vs 79 on 0.1266), while the men get more weight on the lower pass rate (1312 on 0.1905 vs 1259 on 0.2629). If the weights are not constrained, the lower bound on the selection ratio in the pooled data is $0.1266/0.2629 \approx 48.16\%$, the numerator being the smaller of the women's two pass rates and the denominator being the larger of the men's two rates. Likewise, the upper bound is $0.1765/0.1905 \approx 92.65\%$. The observed selection ratio of 68.5% is well inside the permissible range.

- § 4. Only (a) is right; (b) has area about 200%, and (c) has the wrong units.
- § 5. About 1 year. One month is way too small (for instance, 68% of the students can hardly be within one month of average); 5 years is too large.
- § 6. False. These data are cross sectional, not longitudinal. The 50-year-old men were born in 1955, and went to school when educational levels were lower than they are now. Furthermore, the material taught was less relevant to today's economy. In these respects, the 60-year-olds are in worse shape. Education is one reason why the curve of age-specific average incomes flattens out. The effect of education is confounded with the effect of age.
- Comment. Other data suggests that for each birth cohort, average income increases fairly steadily with age.

- § 9. (a) The 70th percentile is about 0.52 in standard units, or 60 in original units. The 80th percentile is about 0.84 in standard units, or 67 in original units. The brothers are separated by about 7 points.

(b) The sisters are separated by about 9 points.

Comment. As you go further into the tails of the normal distribution, each extra percentile covers a greater and greater distance in the original units—because the normal curve gets lower and lower.

- § 12. (a) True.
- (b) True. The correlation between y and x equals the correlation between x and y .
- (c) False, as (a) and (b) should demonstrate.
- (d) False.

Answers to Review Exercises: Chapter 16

1. Option (iii) is the best. The chance error is not likely to be 0 exactly, but should be small relative to the number of draws.
3. Both are wrong. Luck and the law of averages have nothing to do with it—the chances stay the same, every time.
4. (a) 60 rolls. With more rolls, the percentage of aces will be closer to $16\frac{2}{3}\%$. You want the percentage of aces to be far from $16\frac{2}{3}\%$. Chance error in the percentages is working against you, choose the smaller number of rolls.
(b) 600 rolls. Now chance error in the percentages is working for you, choose the larger number of rolls.
(c) 600 rolls. Like (b).
(d) 60 rolls. As you roll more and more, there get to be more and more possibilities, no particular one can be very likely. Take a more extreme case: with 6000 rolls, you can get 1000 aces, or 1001, or 1002, There are lots of possibilities, each one individually has a small chance.
6. Possibility (i) is better. Reason: $10/15 = 20/30 = 2/3$. Option (i) is like tossing a coin 15 times, and asking for $2/3$ or more heads. Option (ii) ups the number of tosses to 30. With the bigger number of tosses, you are less likely to get $2/3$ or more heads. This is like exercise 4(a).
7. The score is like the sum of 25 draws from the box $\boxed{4 \ -1 \ -1 \ -1 \ -1}$.
9. Choose (ii). With more draws, the percentage of reds is likely to be closer to the percentage in the box, therefore, above 50%. (When the percentage of reds among the draws is above 50%, more reds are drawn than blues, and you win the dollar.) Exercise 9 is like exercise 4(b) or 6.
10. (a) $30/200 = 0.15$. (b) -0.1 . (c) Average = sum/200.
(d) The same: $5/200 = 0.025$, so the options describe the same event in different language.

4. The number of aces in 180 rolls of a die is like the sum of 180 draws from a box with 1 ticket marked "1" and 5 tickets marked "0." The number of aces will be around 30, give or take 5 or so. There is about a 99.7% chance that the number of aces will be in the range 15 to 45. About 99.7% of the people should get a number in that range.

7. (a) $321/100 = 3.21$. (b) $3.78 \times 100 = 378$.

- (c) The average of the draws will be between 3 and 4 when the sum is between 300 and 400. The expected value for the sum is 350, and the SE is

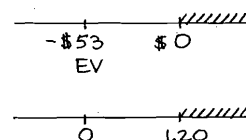
$$\sqrt{100} \times 1.7 = 17,$$

so the chance is about 99.7%.

9. (a) is false, (b) and (c) are true. The reason: if you play (i), the net gain is like the sum of 1000 draws from a box with 12 tickets marked \$2 and 26 marked -\$1.

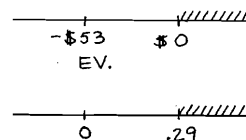
The average of the the box is $-\$2/38$ and the SD is about \$1.39. The expected value for the net gain is $-\$53$ and the SE is \$44. If you play (ii), there is another box, with a bigger SD. The net gain has the same expected value but the SE is \$182. Chance variability helps you overcome the negative expected value, so you are more likely to come out ahead with B. Chance variability also makes it more likely that you will lose big.

The chance of coming out ahead with (i) is about 12%.



Chance $\approx$ shaded area $\approx 12\%$.

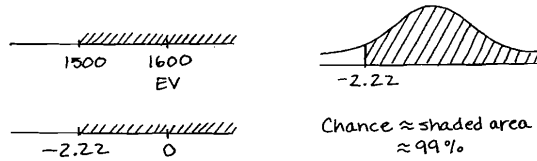
The chance of coming out ahead with (ii) is about 38%.



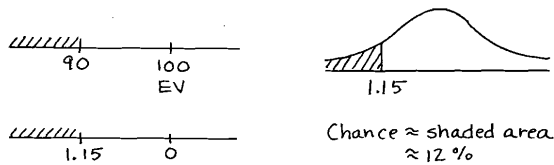
Chance $\approx$ shaded area $\approx 38\%$.

11. You have to change the box, to $\boxed{00013}$. The average of the box is 0.8, and the SD is about 1.2. So the sum will be around 80, give or take 12 or so.

2. (a) The average of the box is 4 and the SD is about 2.24. The EV for the sum is 1600 and the SE is about $\sqrt{400} \times 2.24 \approx 45$. The chance is about 99%.



- (b) The number of 3's is like the sum of 400 draws from the box $[0 \ 1 \ 0 \ 0]$. The expected number is 100 and the SE is 8.66. The chance is about 12%.

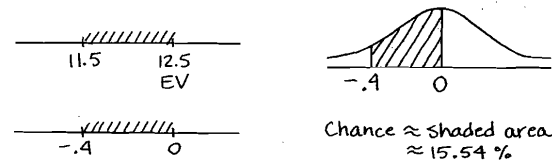


Comment. For more accuracy, use the continuity correction on part (b),

4. If you get 12 heads, you automatically get 13 tails. The chance of getting 12 heads can be found using the method of chapter 15:

$$\binom{25}{12} \times \left(\frac{1}{2}\right)^{25} = \frac{5,200,300}{33,554,432} \approx 15.50\%$$

Or, use the method of example 1 on p.317. The expected number of heads is 12.5, and the SE is 2.5. The normal curve gives a very good approximation:

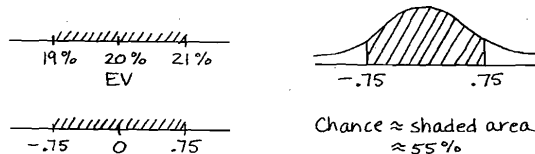


6. Something is wrong with COIN. With a million tosses, the number of heads should be around 500,000 give or take 500 or so. COIN is 4 SEs too high. (The SE for the number of heads is a tiny fraction of the number of tosses.)
9. (a) is true, (b) is false, you don't have enough draws to use the normal approximation. See exercise 6, p.324.
- Technical comment. If you use the continuity correction, the normal approximation is not ridiculous—for this symmetric interval.

Answers to Review Exercises: Chapter 19

5. This estimate is likely to be too high. With smaller households, the interviewer is less likely to find someone at home. So the survey procedure is, on average, replacing smaller households by larger ones.
6. No. Different sorts of students are more likely or less likely to walk through different parts of the campus at different times, and the chances are difficult or impossible to figure (p. 341).
9. The smaller one. As the size of the hospital goes up, the percentage of male births gets closer to 52%, and is less likely to exceed 55%. See section 16.1.
12. (a) Only the “high achievers” are listed in the sampling frame.
(b) There seems to be quite a lot of room for bias here. How were the 5,000 students selected? What about non-response bias?

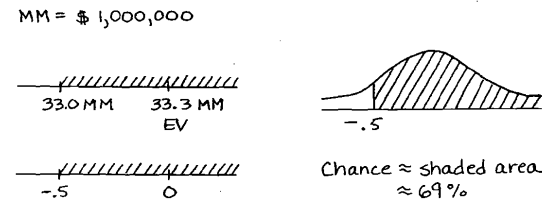
3. (a) There should be 50,000 tickets in the box. The box is the population—all 50,000 forms.
 (b) Each ticket shows a 0 (gross income under \$50,000) or a 1 (gross income over \$50,000).
 (c) False: the SD of the box is $\sqrt{0.2 \times 0.8} = 0.4$.
 (d) True. The draws are the sample.
 (e) The number of sample forms with gross incomes over \$50,000 is like the sum of 900 draws from the box. The expected value for the sum is 180. The SE for the sum is $\sqrt{900} \times 0.4 = 12$. The number of sample forms with gross incomes over \$50,000 will be around 180, give or take 12 or so. Now 12 out of 900 is about 1.33%. The percentage of forms in the sample with gross incomes over \$50,000 will be 20.00%, give or take 1.33% or so. The chance is about 55%—



- (f) Can't be done with the information given. You need to know the percentage of forms with gross incomes over \$75,000. And you can't use the normal curve, because these data are far from normal.

4. (a) 50,000. (b) A gross income. (c) True. (d) True.

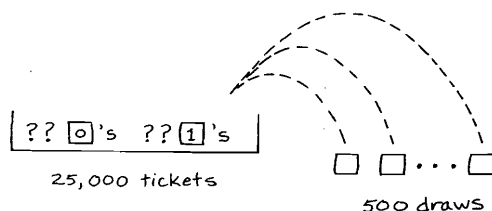
The total gross income is like the sum of 900 draws from the box, whose average is \$37,000. The SD is \$20,000. The expected value for the total is $900 \times \$37,000 = \$33,300,000$. The SE is $\sqrt{900} \times \$20,000 = \$600,000$. The chance is about 69%.



In exercise 3, you were classifying and counting, so you had to change the box; here, you are working with a sum, so it would be a bad idea to change the box.

6. Option (ii) is right. In absolute terms, the California sample will be much bigger, and therefore more accurate for estimating percents; the size of the sample relative to the size of the state is basically irrelevant (section 4).
 7. (a) is true and (b) is false. The expected value is computed with no chance error: the EV equals the percentage of 1's in the box.
 (c) is true and (d) is false. The percentage of 1's among the draws will be off its expected value, due to chance error. The SE for the number of 1's among the draws is $\sqrt{500} \times \sqrt{0.25 \times 0.75} \approx 9.7$. The SE for the percentage of 1's among the draws is $(9.7/500) \times 100\% \approx 2\%$.
 (e) is true and (f) is false: you know what's in the box.

2. (a) The sample is like 500 draws from a box with 25,000 tickets; each ticket is marked 1 (has a computer) or 0 (does not have a computer). The number of sample households with computers is like the sum of the draws.



The fraction of 1's in the box is unknown, but can be estimated by the fraction in the sample, as $239/500 = 0.478$. On this basis, the SD of the box is estimated as $\sqrt{0.478 \times 0.522} \approx 0.50$. The SE for the number of sample households with computers is estimated as $\sqrt{500} \times 0.50 \approx 11$, and 11 out of 500 is 2.2%. The percentage of households in the town with internet access is estimated as 47.8%, and the estimate is likely to be off by 2.2% or so.

- (b) The 95%-confidence interval is $47.8\% \pm 4.4\%$.

Comment. You need to estimate the percentage for the town from data for the town: the national figures may not apply. (In this case, the town seems pretty close to the national average.)

3. (a) 1.4%, 0.5 of 1%.

- (b) Hard to do: the box is lopsided and the normal approximation won't work very well. (See exercises 5–6 on p. 324; exercises 3–4 on p. 383.)

5. (a) The box has millions of tickets, one for each 17-year-old in school that year. Tickets are marked 1 for those who knew that Chaucer wrote *The Canterbury Tales*, and 0 for the others. The data are like 6000 draws from the box, and the number of students in the sample who know the answer is like the sum of the draws. The fraction of 1's in the box can be estimated from the sample as 0.361. On this basis, the SD of the box is estimated as $\sqrt{0.361 \times 0.639} \approx 0.48$. The SE for the number of students in the sample who know the answer is estimated as $\sqrt{6000} \times 0.48 \approx 37$. The SE for the percentage is $37/6000$, which is about 0.6 of 1%. The percentage of students in the population who know the answer is estimated as 36.1%, give or take 0.6 of 1% or so. The 95%-confidence interval is $36.1\% \pm 1.2\%$.

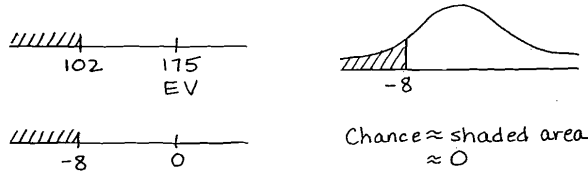
- (b) $95.2\% \pm 0.6$ of 1%.

12. Option (ii) is it. For example, about 95% of the estimates will be right to within 2 SEs, about 99.7% of them will be right to within 3 SEs, and so forth.

2. (a) True. That's what the SE does for you (pp.417, 422).
 (b) True. The 68%-confidence interval is "sample average $\pm$ SE for average."
 (c) False (p.417).
 3. Model: there is a box with 50,000 tickets, one for each household in the town. The ticket shows the commute distance for the head of household. The data are like 1000 draws from the box. The SD of the box is unknown, but can be estimated by the SD of the data, as 9.0 miles. The SE for the sum of the draws is estimated as $\sqrt{1000} \times 9 \approx 285$ miles, and the SE for the average is estimated as $285/1000 \approx 0.3$ miles.
 (a) The average commute distance of all 50,000 heads of households in the town is estimated as 8.7 miles; this estimate is likely to be off by 0.3 miles or so.
 (b) $8.7 \text{ miles} \pm 0.6 \text{ miles}$.
- Comment. The normal curve is used to approximate the probability histogram for the sample average, not the histogram for the data; the data are skewed, with a long right hand tail.
6. (a) Can't be done, you would need to use the half sample method (section 22.5).
 (b) $SE \approx 1$.
 8. (a) True: the interval is "average $\pm$ SE."
 (b) True: section 21.3.
 (c) The data don't follow the normal curve, but the 68% might be right; you need the data to tell. (To see that the data don't follow the curve: enrollments can't be negative, but the SD was a lot bigger than the average, so there must have been a long right hand tail.)
 (d) False: 325 is not the SD. (The data aren't normal, which is another problem.)
 (e) False. The normal curve is being used on the probability histogram for the sample average, not the data (pp.411 and 418-19).
 9. 1.7 ± 0.1 .

10. (a) True: the SE is estimated from the sample data, as on p.416.
 (b) False. There is no such thing as a 95%-confidence interval for the *sample average*, you know the sample average. It's the population average that you have to worry about (pp.385-86).
 (c) True.
 (d) False. This confuses the SD with the SE. And it's ridiculous, because a household must have a whole number of persons (1, or 2, or 3, and so forth).
 (e) False. For instance, if household size followed the normal curve, there would be many households with a negative number of occupants; we're not ready for that.
 (f) True. See pp.411 and 418-19. Even though household size does not follow the normal curve, you can still use the normal curve to approximate the probability histogram for the sample average.
11. There is too much spread in the histogram: the SE for the average is only about 0.3, there is a lot of area outside the range $EV \pm 3 SE$.
12. This is not a 95%-confidence interval: the class is a sample of convenience, not a probability sample.

1. (a) True (p.479).
(b) False. The null says it's chance, the alternative says it's real (pp.477-78).
3. Null hypothesis: the data are like 200 draws from the box $\boxed{1 \ 1 \ 1 \ 0}$, with 1 = blue and 0 = white. The expected number of blues is 150, and the SE is about 6, so $z = (\text{obs} - \text{exp})/\text{SE} = (142 - 150)/6 \approx -1.3$ and $P \approx 10\%$. This is marginal, could be chance.
5. The box has one ticket for each freshman at the university, showing how many hours per week that student spends at parties. So there are about 3000 tickets in the box. The data are like 100 draws from the box. The null hypothesis says that the average of the box is 7.5 hours. The alternative says that the average is less than 7.5 hours. The observed value for the sample average is 6.6 hours. The SD of the box is not known, but can be estimated from the data as 9 hours. On this basis, the SE for the sample average is estimated as 0.9 hours. Then $z = (\text{obs} - \text{exp})/\text{SE} \approx (6.6 - 7.5)/0.9 = -1$. The difference looks like chance.
6. (a) In the best case for Judge Ford, we are tossing a coin 350 times, and asking for the chance of getting 102 heads or fewer. The expected number of heads is 175 and the SE is a little over 9, so $z \approx (102 - 175)/9 \approx -8$. The chance is about 0.

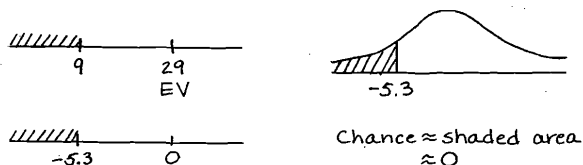


- (b) 100 draws are made at random without replacement from a box with 102 1's and 248 0's, where 1 = woman and 0 = man. The expected number of 1's is 29. If the draws are made with replacement, the SE for the number of 1's is $\sqrt{100 \times \sqrt{0.29 \times 0.71}} \approx 4.54$. The correction factor is

$$\sqrt{\frac{350 - 100}{350 - 1}} \approx 0.846.$$

The SE for the number, when drawing without replacement, is $0.846 \times 4.54 \approx 3.8$.

So $z \approx (9 - 29)/3.8 \approx -5.3$. The chance is about 0.



- (c) Judge Ford was not choosing at random; he was excluding women.

Comment. After Hans Zeisel made a statistical analysis of Judge Ford's procedures, the percentage of women jurors went up quite dramatically; see note 15 to chapter 26.

8. Model: there is one ticket in the box for each person in the county, age 18 and over. The ticket shows that person's educational level. The data are like 1000 draws from the box.

Null: the average of the box is 13 years.

Alt: the average of the box isn't 13 years.

The expected value for the average of the draws is 13 years, based on the null. The SD of the box is unknown (there is no reason the spread in the county should equal the spread in the nation), but can be estimated as 5 years—the SD of the data. On this basis, the SE for the sample average is estimated as 0.16 years. The observed value for the sample average is 14 years, so

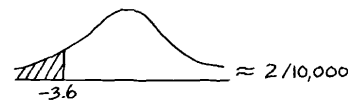
$$z = (\text{obs} - \text{exp})/\text{SE} = (14 - 13)/0.16 \approx 6,$$

and $P \approx 0$. This is probably a rich, suburban county, where the educational level would be higher than average.

10. Null: the 3 Sunday numbers are like 3 draws made at random (without replacement) from a box containing all 25 numbers in the table. The average of these numbers is nearly 436, and their SD is just about 40. The EV for the average is 436, and the SE is 22; we are using the correction factor here. The 3 Sunday numbers average about 357, so

$$z = (\text{obs} - \text{exp})/\text{SE} = (357 - 436)/22 \approx -3.6,$$

and $P \approx 2/10,000$.



Comments. (i) Many deliveries are induced, and some are surgical, so obstetricians really can influence the timing.

(ii) In all, there are $25!/(3! \times 22!) = 2300$ samples of size 3. The 3 Sunday numbers are 344, 377, 351. The number for Saturday, August 6, is also 377; and all the other numbers in the table are larger. Therefore, exactly 2 samples have an average equal to the Sunday average, and no sample average is smaller. The exact significance probability is $2/2300 \approx 9/10,000$, compared to $2/10,000$ from the curve.

12. (a) There are 59 pairs, and in 52 of them, the treatment animal has a heavier cortex. On the null hypothesis, the expected number is $59 \times 0.5 = 29.5$ and the SE is $\sqrt{59} \times 0.5 \approx 3.84$. So 52 is nearly 6 SEs above average, and the chance is close to 0. Inference: treatment made the cortex weigh more.

1. No. The expected number of positives is 250, and the SE is $\sqrt{500} \times 0.5 \approx 11$. The observed number is 2.4 SEs above the expected. (Here, a one-sample test is appropriate.)

3. There are two samples, you need to make a two-sample z-test. Model: there are two boxes. The 2005 box has a ticket for each person in the population, marked 1 for those who would rate clergymen "very high or high," and 0 otherwise. The 2005 data are like 1000 draws from the 1985 box. The 2000 box is set up the same way. The null hypothesis says that the percentage of 1's in the 2005 box is the same as in the 2000 box. The alternative hypothesis says that the percentage of 1's in the 2005 box is smaller than the percentage of 1's in the 2000 box.

The SD of the 2005 box is estimated from the data as $\sqrt{0.54 \times 0.46} \approx 0.50$. On this basis, the SE for the 2005 number is $\sqrt{1000} \times 0.50 \approx 16$: the number of respondents in the sample who rate clergymen "very high or high" is 540, and the chance error in that number is around 16. Convert the 15 to percent, relative

to 1000. The SE for the 2005 percentage is estimated as 1.6%. Similarly, the SE for the 2000 percentage is about 1.5%.

The SE for the difference is computed from the square root law (p. 502) as

$$\sqrt{1.6^2 + 1.5^2} \approx 2.2\%.$$

The observed difference is $54 - 60 = -6\%$. On the null hypothesis, the expected difference is 0%. So $z = (\text{obs} - \text{exp})/\text{SE} = -6/2.2 \approx -2.7$, and $P \approx 3/1000$. The difference is real. What the cause is, the test cannot say.

Comment. Either a one-sided or a two-sided test can be used. Here, the distinction is not so relevant: for discussion, see chapter 29.

7. (a) This is like the radiation-surgery example in section 4. (Also see review exercises 5 and 6 above.) The SE for the treatment percent is 2.0%. The SE for the control percent is 4.0%. The SE for the difference is 4.5%. The observed difference, 1.1%, is only 0.24 of an SE. This could easily be due to chance. Income support was a good idea that didn't work.
- (b) This is like example 4. The SE for the treatment average is 0.7 weeks; for the control average, 1.4 weeks; for the difference, 1.6 weeks. The observed difference is -7.5 weeks. So $z = -7.5/1.6 \approx -4.7$ and $P \approx 0$. The difference is real. Income support makes the released prisoners work less, which might explain the findings in part (a).
8. The data can be summarized as follows:

	Prediction Request	Request only
Predicts	22/46	NA
Agrees	14/46	2/46

- (a) This is like the radiation-surgery example in section 4. The two percentages are 47.8% and 4.4%. The SEs are about 7.4% and 3.0%, respectively. The difference is 43.4% and the SE for the difference is 8%. So $z = 43.4/8 \approx 5.4$ and $P \approx 0$. The difference is real. People overestimate their willingness to do volunteer work.
- (b) The two percentages are 30.4% and 4.4%. The SEs are about 6.8% and 3.0%, respectively. The difference is 26% and the SE for the difference is 7.4%. So $z = 26/7.4 = 3.5$ and $P \approx 2/10,000$. The difference is real. Asking people to predict their behavior changes what they will do.
- (c) Here, a two-sample z-test is not legitimate. There is only one sample, and two responses for each person in the sample. Both responses are observed, so the method of section 4 does not apply. The responses are correlated, so the method of example 3 does not apply. See p. 517.

Comment. In parts (a) and (b), the number of draws is small relative to the number of tickets in the box. So there is little difference between drawing with or without replacement, and little dependence between the treatment and control averages. See pp. 510, 517.

10. This test is not legitimate. There is dependence between the first-borns and second-borns.

1. (a) (i) (b) (iii). See exercises 3–6 on pp. 539–40.

2. Use the method of sections 1–2.

Observed	Expected
1	17.6
10	30.1
16	7.4
35	6.9

$\chi^2 \approx 150$ on 3 degrees of freedom, so $P \approx 0$ and option (ii) is right.

Comments. (i) Judges prefer well-educated grand jurors.

(ii) The expecteds do not have to be whole numbers. For instance, if you roll a die 100 times, the expected number of aces is 16.666

3. Use the method of section 4. The expecteds are as follows (row and column sums are a bit off due to rounding).

	Married	Widowed, divorced, or separated	Never married
Employed	772.0	103.3	221.7
Unemployed	66.2	8.9	19.0
Not in labor force	28.9	3.9	8.3

$\chi^2 \approx 14.2$ on 4 degrees of freedom, which is off the end of the table. By computer, $P \approx 0.7$ of 1%. This is not chance variation. The married men do better at getting jobs. (Or, men with jobs do better at getting married: the χ^2 -test will not tell you which is the cause and which is the effect.)

7. Use the method of sections 1–2: $\chi^2 \approx 0.2$ on 2 degrees of freedom, $P \approx 90\%$, a good fit.

8. Make a χ^2 -test, as in sections 1–2. We are interested in the chances, not just the average: $\chi^2 \approx 2.6$ on 5 degrees of freedom, $P \approx 75\%$, a good fit.

Answers to Review Exercises: Chapter 29

4. Yes: data snooping (p.547).
5. It is hard to make sense out of “statistical significance” here, because there is no reasonable chance model for the data. The inner planets do not form a sample, they are the inner planets; similarly for the outer ones. (See exercise 2 on pp.558–59; but see note 20 to chapter 29.)
7. The concept of statistical significance does not apply very well, because the data are for the whole population, rather than a sample (p.556). The difference is practically significant. The center of population is shifting to the West, and that makes a lot of difference to the economy and to the political balance of the country.
9. (a) The question makes sense, and the difference in attitudes is important. (This is a practical judgment, not a statistical one.)
(b) The question makes sense, because the data are based on probability samples; but to answer it, you need to use the half-sample method (section 22.5).
(c) Now this is like example 3 on p.505. The SE for the 2000 percentage is 1.4%; for 1970, the SE is 1.5%. The difference is 38%, and the SE for the difference is 2.1%, so $z = 38/2.1 \approx 18$ and $P \approx 0$.